

DASC7011 Statistical Inference for Data Science
Master of Data Science, HKU

Assignment 1

(Due on September 22, 2026)

1. Let $\mathbf{X} = \{X_n : n \geq 1\}$ and $\mathbf{Y} = \{Y_n : n \geq 1\}$ be two sequences of random variables. It is known that

$$X_n \rightarrow X \quad \text{and} \quad Y_n \rightarrow Y \quad \text{a.s.,}$$

$\mathbb{P}(Y = 0) = 0$, and $\mathbb{P}(Y_n = 0) = 0$ for all n . Show that $Z_n = \frac{X_n}{Y_n}$ converge to $Z = \frac{X}{Y}$ almost surely. **[10 marks]**

2. Let $\{X_i : 1 \leq i \leq n\}$ be a random sample from a population $X \sim F(\mu, \sigma^2)$, a distributional model with finite but unknown mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$. Let

$$\hat{\mu} = \sum_{i=1}^n w_i X_i$$

be a general linear estimator of μ , where w_i 's are (general) weights.

- (a) Show that $\hat{\mu}$ is unbiased if and only if $\sum_{i=1}^n w_i = 1$.
(b) Show that $\hat{\mu}$ is the Best Linear Unbiased Estimator (BLUE, the one with minimal variance among all linear unbiased estimators) if and only if $w_i \equiv \frac{1}{n}$ for all i .

[10 marks]

3. Let $\mathbf{X} = (X_1, \dots, X_n)'$ be a random sample from a population $X \sim F(\mu, \sigma^2)$ with finite but unknown mean $\mu > 0$ and variance $\sigma^2 > 0$. Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Show that $\hat{\sigma}^2$ is a biased estimator of σ^2 . Find its bias.

[10 marks]

4. Read the iris data into R and define the Sepal Length of Setosa and Versicolor as variables X and Y respectively. Refer to the following R codes.

```
data("iris")
X <- iris[iris[, 5] == "setosa", 1]
Y <- iris[iris[, 5] == "versicolor", 1]
```

Denote the mean and variance of X and Y as μ_x , μ_y , σ_x^2 and σ_y^2 , respectively. Do the following tests at the 5% significance level.

- (a) Test the normality of X .
- (b) Assuming $\sigma_x = \sigma_y$, test $\mu_x = \mu_y$ against $\mu_x \neq \mu_y$.
- (c) Assuming $\sigma_x \neq \sigma_y$, test $\mu_x \leq \mu_y$ against $\mu_x > \mu_y$.
- (d) Test $\sigma_x^2 \geq \sigma_y^2/2$ against $\sigma_x^2 < \sigma_y^2/2$.

[20 marks]